

EN 622: Transport Phenomena for Energy Systems

Course Code	EN 622
Course Name	Transport Phenomena for Energy Systems
L-T-P-C	2-0-2-3
Prerequisites	Instructor's consent
Intended for	M. Tech / UG/MS/PhD
Distribution	Core for M. Tech, elective for / UG/MS/PhD
Semester	Odd/Even
Approval	9th Senate (May 2016), in the M.Tech Energy Engineering (Mechanical Design) curriculum

Preamble

The objective of this course is to acquaint the student with important topics on advanced transport phenomena (momentum, heat and mass transport). Topics include laminar and turbulent flow, thermal transport and the energy equation, molecular mass transport including diffusion with homogenous and heterogeneous reactions. Key focus of this course will be to develop physical understanding of the principles discussed and with emphasis on energy applications.

Course Outline

The objective of the course is to introduce the students to topics transport phenomena. The course aims to provide the students an understanding of the fundamental concepts and techniques that are important in the momentum, mass and heat transfer phenomena. Understanding the fundamentals will help in design of various components of the process industries.

Course Modules

- **Module 1:** Introduction to transport phenomena, Vectors and tensors, types/uses of control volume, notion of conservation principles and constitutive relations (4 hours)
- **Module 2:** Mechanisms of momentum transport, shell-momentum balance, equations of change for isothermal systems, applications of shell-momentum balance and equations of change to solve 1D problems in laminar flow, time dependent flow of Newtonian fluids, stream function and velocity potential, time smoothed equations for turbulent flows, boundary layer flow past different bodies (16 contact hours)
- **Module 3:** Thermal conductivity and energy transport, Fourier's law of heat conduction, shell-energy balance and boundary conditions, equations of change for non-isothermal systems, temperature distributions and unsteady heat conduction in solids, steady and unsteady free convection, temperature distributions in turbulent flow and turbulent heat flux, forced convection in tubes, submerged bodies and packed beds (14 contact hours)
- **Module 4:** Mechanisms of mass transport, Fick's law of diffusion, shell-mass balance, equations of change for multi-component systems, laminar flow concentration profiles, steady and unsteady convective mass transport, diffusion in gases and liquids, multi-component mass diffusion, reaction rates, mechanisms, time scales (8 contact hours)

References

1. *Transport phenomena*, R. Byron Bird, Warren E. Stewart, Edwin N. Lightfoot, 2nd edition, Wiley, 2001.
2. *Advanced transport phenomena*, J.C. Slattery, Cambridge University Press, New York, 1999.
3. *Fundamentals of momentum, heat and mass transport*, Welty J.R, Wicks J.E, Wilson R.E, Rorrer G, 4th edition, Wiley, 2001.